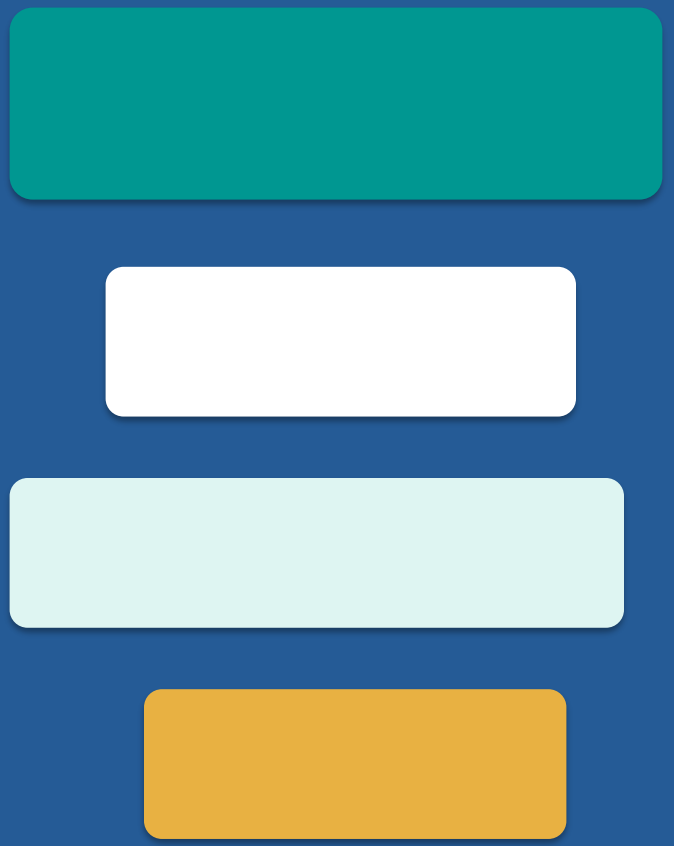
BCA ANALYTICS • PROJECT PROPOSAL

PERSONAL FINANCE ASSISTANT

A goal-based money planning system for students, freelancers, and first-time earners.

A modular application developed across three semesters



TRACK • PLAN • LEARN • IMPROVE

Prepared for academic discussion • Project scope can be refined after user research

Project at a glance

One application, six connected modules, three semesters.

Target users

Students, freelancers, and recent graduates managing their first or early income.

Core purpose

Turn income, expenses, savings, and goals into a clear, practical financial plan.

Project approach

A simple, explainable, data-driven system—not a trading platform or a promise of returns.

What the application will help users do

Record income and expenses and understand their available balance.

Identify spending patterns and set realistic budgets.

Plan savings toward specific goals and learn about financial products.

Receive transparent, rule-based next-step guidance and monitor progress.

Problem statement

Focus on a specific user need—not the claim that no similar app exists.

Students, freelancers, and recent graduates may have limited experience turning their income, expenses, savings, and future goals into a practical financial plan. While budgeting tools and financial information are available, users may still find it difficult to decide what to prioritise and how their choices relate to their circumstances.

The gap this project will investigate

Information is often spread across multiple tools.

Beginners may not know what to do first.

Generic advice may not reflect cash flow or goal timelines.

Research question

Can a simple, connected planning tool help first-time earners make clearer day-to-day financial decisions?

Target users and their needs

Initial audience can be validated through short interviews or a survey.

College students

Possible income: allowance, stipend, part-time work.

Need: control spending and save for near-term costs.

Freelancers

Possible income: irregular project payments.

Need: plan around fluctuating monthly cash flow.

First-job earners

Possible income: monthly salary.

Need: balance expenses, debt, savings, and longer-term goals.

Validation before development

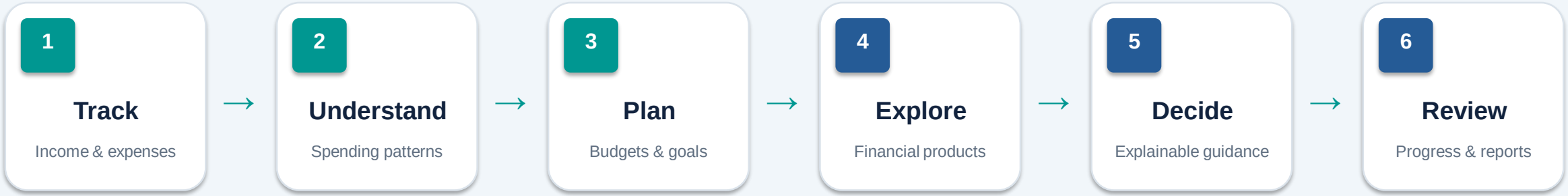
Interview 20–30 people in the target group about real financial decisions they have faced.

Ask what tools they currently use, what is confusing, and what they still cannot do easily.

Use findings to refine the scope; do not assume every target user has the same problem.

Proposed solution and objectives

A connected workflow from tracking to planning and learning.



Project objectives

Build a usable transaction and budgeting tool.

Make personal finance concepts easier to understand.

Help users compare options against goals and constraints.

What success looks like

Calculations are correct for test cases.

Users can complete core tasks without help.

Guidance is understandable and explains its assumptions.

Six-module architecture

Modules are connected, but each has a clear deliverable.

01

Income & Expense Tracker

Record, categorize, edit, and delete transactions.

02

Budget & Spending Analytics

Summarize spending, compare budgets, show patterns.

03

Savings & Goal Planner

Set goals, deadlines, and monthly savings targets.

04

Financial Learning & Product Explorer

Explain product types, risks, liquidity, and time horizons.

05

Rule-Based Financial Guidance

Prioritize next steps using transparent rules and user inputs.

06

Progress, Reports & Review

Track progress, export summaries, and review monthly trends.

Module 1: Income & Expense Tracker

Planned for Semester 1

Purpose: give users one place to record money coming in and going out.

Core features

- Add income and expense entries.
- Choose category, date, amount, and note.
- Edit or delete a transaction.
- View transaction history and running totals.

Inputs

- Amount and transaction type
- Category and date
- Optional description

Outputs

- Total income
- Total expenses
- Available balance
- Transaction list

SUCCESS CHECK For a defined test dataset, totals and balance match expected calculations exactly.

Module 2: Budget & Spending Analytics

Planned for Semester 1

Purpose: help users understand spending patterns and compare actual spending with a plan.

Core features

- Set monthly budget limits by category.
- Show actual spending against budget.
- Display category breakdowns and simple charts.
- Flag categories approaching or exceeding limits.

Inputs

- Transactions from Module 1
- Monthly budget amounts
- Selected month

Outputs

- Budget vs actual
- Category summaries
- Overspending alerts
- Savings-rate estimate

SUCCESS CHECK Users can identify their largest expense categories and see where they exceeded a budget.

Module 3: Savings & Goal Planner

Planned for Semester 2

Purpose: turn a goal into a measurable savings plan with a target and timeline.

Core features

- Create a goal and target amount.
- Set a target date and current savings.
- Calculate the remaining amount and monthly target.
- Track progress as savings are added.

Inputs

- Goal name and amount
- Current savings
- Target date

Outputs

- Amount remaining
- Required monthly saving
- Progress indicator

SUCCESS CHECK

For a target of ₹12,000 in 6 months with ₹0 saved, the basic target is ₹2,000 per month.

Module 4: Financial Learning & Product Explorer

Planned for Semester 2

Purpose: explain common financial products in beginner-friendly language.

Core features

- Learning cards for savings accounts, FD, RD, and mutual funds.
- Compare risk, liquidity, time horizon, and access to money.
- Explain terms such as inflation, compounding, SIP, and diversification.
- Use sourced, date-stamped information where product data is shown.

Inputs

- User's learning topic
- Goal time horizon
- General product information

Outputs

- Plain-language explanations
- Comparison table
- Risk and liquidity notes

SUCCESS CHECK Users can explain the basic differences between selected products after using the learning section.

Module 5: Rule-Based Financial Guidance

Planned for Semester 3

Purpose: suggest sensible next steps based on clear, explainable rules—not opaque AI predictions.

Core features

- Check cash flow and essential expenses first.
- Consider debt obligations and accessible savings.
- Match goals with time horizon, liquidity, and risk preference.
- Explain why a next step is shown and what assumptions were used.

Inputs

- Income and expenses
- Debt and savings status
- Goals and time horizon
- Risk preference

Outputs

- Prioritized next steps
- Reasoning and assumptions
- Educational options to explore

SUCCESS CHECK Every recommendation can be traced to a documented rule and is framed as educational guidance, not guaranteed investment advice.

Module 6: Progress, Reports & Review

Planned for Semester 3

Purpose: help users review changes over time and stay engaged with their plan.

Core features

- Monthly summaries of income, expenses, and savings.
- Compare this month with prior periods.
- Show goal progress and budget trends.
- Export a report as CSV or PDF; optional reminders.

Inputs

- Stored transactions
- Budgets and goals
- Selected reporting period

Outputs

- Monthly report
- Trend summaries
- Goal progress report
- Exportable data

SUCCESS CHECK

Users can generate a correct monthly report and compare at least two periods using sample data.

Three-semester implementation roadmap

Build a usable core first, then extend it without rewriting the application.

SEMESTER 1

Build the core

Module 1: transaction entry and totals

Module 2: budgets and spending summaries

Deliverable: working prototype with sample data

SEMESTER 2

Add planning and learning

Module 3: savings goals and targets

Module 4: financial education and comparison

Deliverable: goal planner and learning library

SEMESTER 3

Connect and evaluate

Module 5: transparent rule-based guidance

Module 6: reports and progress monitoring

Deliverable: integrated app, testing, and final report

At the end of every semester: demo working features, document test results, and gather user feedback.

Suggested technology stack

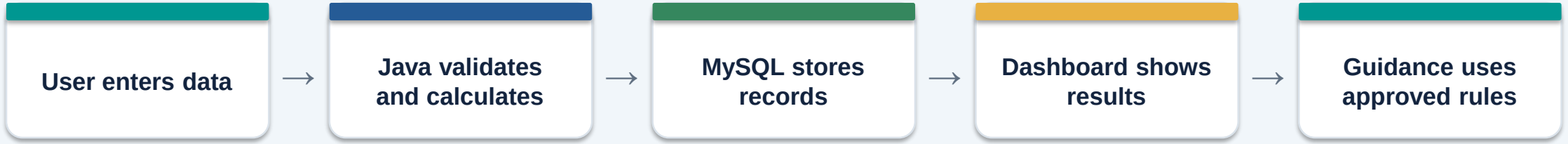
A practical stack for a Java-based web application.

Frontend	**HTML + CSS**	Forms, layouts, dashboard, transaction list.
Backend	**Java + Spring Boot**	Business rules, validation, calculations, request handling.
Database	**MySQL**	Users, transactions, budgets, goals, and reports.
Charts	**Chart.js or server-rendered summaries**	Visualize category totals and monthly trends when needed.
Tools	**IntelliJ IDEA / VS Code + Git**	Development, debugging, version control, and collaboration.

Keep the first version small: build the tracker and budget logic before adding external APIs, bank connections, or AI.

How the system works

High-level data flow and core records.



Core data entities

User: account/profile settings (later phase).

Transaction: type, amount, category, date, note.

Budget: category, month, planned limit.

Goal: target amount, current savings, target date.

Design principles

Validate amounts and dates before saving.

Use decimal-safe monetary values in the backend.

Protect user data; collect only what is necessary.

Make calculations and guidance explainable.

Testing and evaluation plan

Evaluate whether the app works correctly and is understandable—not just whether it runs.

Functional tests

- Add, edit, and delete transactions
- Validate zero, negative, and blank amounts
- Check income, expense, and balance totals
- Verify monthly filters

Usability tests

- Ask target users to complete core tasks
- Record where they get confused
- Check if categories and labels are clear
- Collect improvement suggestions

Outcome measures

- Calculation accuracy on test cases
- Task completion rate
- Time to enter a transaction
- User understanding of guidance

Example acceptance criteria

All predefined calculation tests pass.

A user can add and review a transaction without developer assistance.

The app clearly labels sample assumptions and never presents estimated returns as guaranteed.

Limitations, risks, and responsible design

A credible project is clear about what it will—and will not—do.

Data privacy

Use sample data during development; avoid collecting bank credentials or unnecessary sensitive information. Add authentication and access controls before multi-user deployment.

Financial accuracy

Use reliable sources for educational content, record the date of product information, and explain assumptions. Rates and product terms can change.

Advice boundaries

The app provides budgeting support and general education. It does not guarantee returns or claim to replace a qualified financial professional.

Manageable scope

Do not include real-time trading, bank-account integrations, payment execution, or black-box AI in the initial build.

The goal is informed decision-making—not telling every user to buy a particular investment.

Expected outcomes

A working core	Transaction tracking and budget analytics by the end of Semester 1.
A connected planner	Goals and beginner-friendly financial learning in Semester 2.
A complete prototype	Explainable guidance, reports, and evaluation by Semester 3.

The project will be judged by its usefulness, clarity, correctness, and evidence from target users—not by claiming the problem is entirely new.

Thank you • Questions and feedback